

## Net change



1

a 40 kangaroos

b 8 months

2

a i No ii Yes iii No iv Yes

b 3000

c i Yes ii Yes iii No iv No

d 36 000

3

a  $-2.02^{\circ}\text{C}/\text{minute}$  b  $-41.74^{\circ}\text{C}$

c  $-4.64^{\circ}\text{C}/\text{minute}$

4

a  $-\frac{24}{\pi}$

b  $-\frac{30}{\pi}$

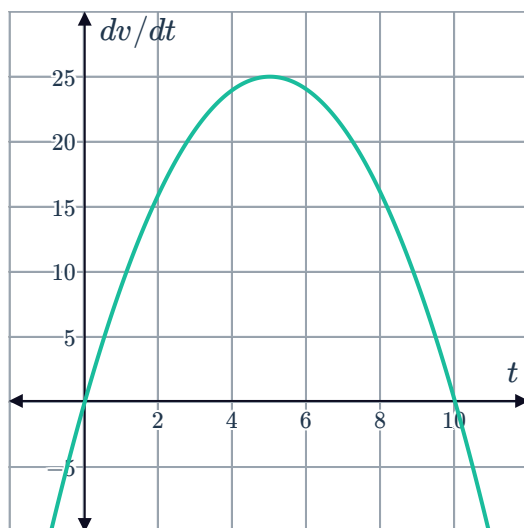
5

a 5818.82 L

b 175.71 L

6

a



b 25 L/min

c  $V = 5t^2 - \frac{t^3}{3}$

d 15 minutes

7

a 3700 tonnes/year b 136 tonnes/year c 100 tonnes/year d 4140 tonnes

8

a  $M = \frac{81t^2}{2} - \frac{t^4}{4} + 1000$

b 2568 kg

c  $t = 9$



9

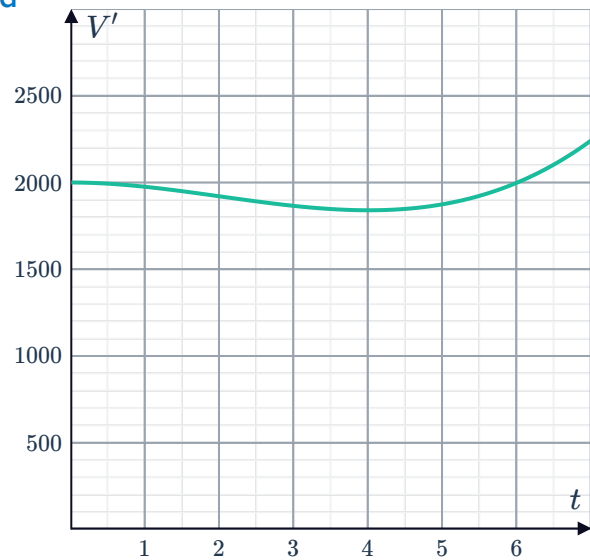
a 2000 L/h

b

| $t$   | -1 | 0 | 1   | 4 | 6   |
|-------|----|---|-----|---|-----|
| $V''$ | 75 | 0 | -45 | 0 | 180 |

c  $t = 7$  at 2245 L/h

d



e 5831.25 units<sup>2</sup>

f The total change in the amount of water in the tank in the first 3 hours is 5831.25 L.

10

a  $\frac{dV}{dt} = -25$

b  $V = 15\,625 - 25t$

c  $t = 625$

11

a  $m = \frac{2t^{\frac{3}{2}}}{360} + \frac{t^3}{4320}$

b  $\frac{176}{135}$  g

c 0.004 g

12

a 145 customers

b 184 customers

c 134 customers

## Cost, revenue and profit applications

13

a \$6222

b \$1037/item



14

**a**  $P(x) = 0.0004x^3 + 0.6x^2 - 4.1x - 60$

**b** \$2730.20

**c** \$9493.40

15

a

**i** 40 000 units<sup>2</sup>      **ii** 32 500 units<sup>2</sup>

**b**    \$40 000

**c** \$34 500

**d** \$5500

e  $-\$5000$

16

**a**  $C = 0.1x^4 + 2x^2 + 2000$

**b** \$632 000

17

**a**  $C(x) = 3x^3 - 35x^2 + 500x + \text{initial costs}$

**b** \$532 000

18

**a**  $C(x) = 240x^{\frac{1}{2}} + \text{initial costs}$

**b** \$960

19

**a**     $-\$17\,601.30$

**b** —\$704 700

**c**  $-\$23\,490/\text{item}$

20

**a**  $P(x) = 3x^2 + 2x^3 + 5x$

**b** \$56 850

21

**a** \$5.03

**b** \$4.57

22

$$\mathbf{a} \quad R = 401t - \frac{250}{(t+1)^2} + 250$$

**b** \$449 611.11/week

**c** \$402 842.40

23

a

i \$81.62

- ii \$9000

- iii \$90

**b**

- i \$31.42

ii \$1607.70

- iii \$16.08

